

Customer Churn Analytics Project:- Project Plan

1. Project Overview

This project aims to analyze customer churn patterns for a subscription based business and uncover the key drivers behind customer attrition. The objective is to build a full analytics pipeline using SQL, Excel, and Tableau that enables stakeholders to understand who is churning, why they are churning, which customers are at highest risk, and how churn impacts revenue.

2. Business Problem Statement

Customer churn is one of the most critical challenges for subscription based businesses. Without clear visibility into churn behavior, organizations struggle to retain customers, forecast revenue, and design effective engagement strategies. This project addresses these gaps by providing structured, data-driven insights into churn patterns and customer behavior.

3. Tools & Technologies

- SQL: Core data transformation, churn logic, cohort creation, risk modeling, and aggregations.
- Excel: Data validation, reshaping and final exports.
- Tableau: Dashboarding, storytelling, and stakeholder facing visualizations.

4. Folder Structure & Data Pipeline

The project follows a structured analytics workflow:

Raw Data → SQL Transformations → Excel Validation → Tableau Dashboards → Insights & Recommendations.

Each stage of the pipeline is mapped to a dedicated project folder to ensure clarity, traceability, and reproducibility.

5. Phase 1: Data Understanding & Scoping

The first phase focused on understanding the dataset, identifying relevant columns, defining churn logic, and determining what business questions could realistically be answered with the available data.

Deliverables:

- Metric definitions
- Business problem scoping
- Data limitations documentation
- Analysis roadmap

6. Phase 2: Data Cleaning & Feature Engineering (SQL)

SQL was used as the primary transformation layer. This phase involved building a clean customer base table, defining churn status, calculating tenure, creating risk features, and generating aggregated tables for trend, segmentation, cohort, and revenue analysis.

Deliverables:

- Clean base tables
- Churn logic implementation
- Risk modeling features
- Cohort logic
- Revenue aggregations

7. Phase 3: Validation & Reshaping (Excel)

Excel was used to validate SQL outputs, perform sanity checks, reshape tables for visualization, and ensure metric correctness. Pivot tables and manual QA steps were used to confirm accuracy.

Deliverables:

- Verified datasets
- Tableau ready exports
- QA documentation

8. Phase 4: Visualization & Dashboarding (Tableau)

Tableau was used to convert analytical outputs into interactive dashboards. Each dashboard was designed to answer a specific business question and avoid information overload.

Dashboards Created:

- Executive Overview
- Trends & Movement
- Risk Segmentation
- Revenue Impact
- Behavioral Signals
- Cohort Analysis

9. Phase 5: Insights & Recommendations

This phase focused on translating analytical outputs into business insights. Patterns were identified, key drivers of churn were summarized, and actionable recommendations were documented.

Deliverables:

- Key findings document
- Business recommendations

- Dashboard interpretation notes
- Interview preparation guide

10. Key Metrics Defined

- Churn Rate
- Active Users
- Churned Users
- At Risk Users
- Retention Rate
- ARPU
- Revenue Lost
- Revenue at Risk

11. Success Criteria

The project is considered successful if:

- Dashboards clearly explain churn behavior
- Risk segments are identifiable
- Revenue impact is quantifiable
- Behavioral signals are interpretable
- Insights are actionable
- The project can be explained clearly in interviews

12. Future Enhancements

Potential future extensions of this project include:

- Predictive churn modeling
- ML-based risk scoring
- Real-time dashboards
- Automated retention alerts
- Experimentation frameworks for churn reduction